

A Visual Landscape for Doi–Peliti Reaction-Network Universality

Placing CRNs on the Puiseux stratification of algebraic Dyson–Schwinger equations

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Abstract

When the Doi–Peliti (DP) path integral of a reaction network is rewritten as a Dyson–Schwinger equation (DSE) for its dressed propagator, one obtains an equation of the form $G(z) = z\phi(G(z))$ with ϕ a polynomial in G . This places the DP generating function squarely inside the class of algebraic generating functions studied by analytic combinatorics, where the coefficient asymptotics $[z^n]G \sim Az_\star^{-n}n^{-\tau}$ are controlled entirely by the Puiseux exponent of the dominant branch of $F(G, z) = G - z\phi(G)$. The Flajolet–Sedgewick transfer theorem and the Banderier–Drmotá dyadic-ceiling result for positive ($\mathbb{N}$ -algebraic) systems together produce an explicit landscape whose strata $\mathcal{C}_k$ carry the exponents $\tau_k = 1 + 1/k$. We describe this landscape in visual form for the cubic case; we place the directed-percolation family, pair coagulation, triplet annihilation and Schlögl II on it; we note the specific points where these families intersect a higher-order stratum; and we observe that because DP generators contain negative vertex weights whenever annihilation channels are present, DP DSEs routinely fall outside the $\mathbb{N}$ -algebraic cone and can in principle reach the strata that Banderier–Drmotá forbid to positive combinatorics. Nothing in this note is new as mathematics. The novelty, such as it is, is cartographic: the landscape clarifies where reaction networks live relative to one another and where one would have to tune a known system to see an anomalous Puiseux exponent. We flag the experimental candidates we are aware of and are explicit about what the framework does and does not say.

1 Why a landscape view

Two separately mature literatures intersect at the Doi–Peliti DSE. On one side, the analytic combinatorics of algebraic generating functions is largely settled: the Flajolet–Sedgewick transfer theorem (Flajolet and Sedgewick, 2009; Flajolet and Odlyzko, 1990) reads the coefficient asymptotics $[z^n]G \sim Az_\star^{-n}n^{-\tau}$ directly off the Puiseux expansion of the dominant singularity, and Banderier–Drmotá (Banderier and Drmotá, 2015) classify which rational exponents $\tau - 1$ are reachable by $\mathbb{N}$ -algebraic (equivalently, context-free) systems. On the other side, the Doi–Peliti path integral (Doi, 1976; Peliti, 1985; Täuber, 2014) and its response-field extension (Martin et al., 1973; Janssen, 1976; De Dominicis, 1976) produce, after the usual recursive dressing of the propagator, an equation of exactly the form $G = z\phi(G)$ with ϕ polynomial in G whose coefficients encode the reaction-network vertex structure.

The intersection is prosaic: *DP DSEs are algebraic generating functions*. What analytic combinatorics knows about such objects therefore applies, verbatim, to reaction networks. This short note documents a visual consequence of that observation. We plot DSE coefficient space, draw in the stratification of branch orders, and read off where physical reaction networks live. The picture is not new as mathematics. It is, we hope, useful as a map.

Two remarks before beginning. First, we make no claim about coupled DSEs (particle + field), where the relevant object is a resultant of two Lagrange equations and the analysis is more involved. Second, the coefficient asymptotics of a DP dressed propagator are properties of a *signed* diagrammatic sum, not of a probability distribution; we discuss this in Section 6 because it is the point on which the landscape has genuine organising power.

2 Background: algebraic GFs and Puiseux strata

Physical intuition. For a physicist the content of the Flajolet–Sedgewick transfer theorem is elementary: the power-law tail exponent τ of any distribution (cluster size, avalanche duration, response function) is the *shape* of its generating function’s dominant singularity. A square-root branch gives $\tau = 3/2$, the directed-percolation value familiar from cluster and avalanche statistics. A cube-root branch gives $\tau = 4/3$; a quartic-root gives $\tau = 5/4$; a $1/k$ -root cusp gives $\tau_k = 1 + 1/k$ via $[z^n]G \sim n^{-\tau_k}$. We call any τ_k with $k \geq 3$ an *anomalous Puiseux exponent* because it signals that the system is outside the directed-percolation universality class—a different critical theory. Universality classes in the analytic-combinatorics language are thus *shapes of singularities*, and the exponents a physicist measures in the lab are read off those shapes by an elementary transfer.

The Banderier–Drmotá theorem (Banderier and Drmotá, 2015) says something structural and surprisingly strong: any *positive* (probabilistic / $\mathbb{N}$ -algebraic) system—one built from non-negative counting weights, no matter how baroque the setup—can only reach Puiseux orders k that are powers of two: $k \in \{2, 4, 8, 16, \dots\}$. Equivalently, only the *dyadic tower* of exponents $\tau \in \{3/2, 5/4, 9/8, 17/16, \dots\}$ is accessible to pure probability; every other τ_k in the ladder— $4/3, 6/5, 7/6, 8/7, 10/9, \dots$ —is *structurally forbidden* to any probabilistic construction. Physically this is a no-go theorem of the same character as Mermin–Wagner: if you ever measure a non-dyadic τ , the underlying object is positively *not* a probability distribution; it has to be a signed structure (interference, cancellation, the kind of thing a path integral with minus signs produces). This is what the phrase “dyadic ceiling” means in the rest of the paper. Doi–Peliti path integrals with annihilation channels are signed by construction—the generator $(z+1)^\ell - (z+1)^k$ with $\ell < k$ contributes negatively—so they are not bound by the ceiling, which is why the anomalous strata become reachable at all.

Let $\phi(y) = 1 + \sum_{j \geq 1} \phi_j y^j$ be a polynomial of degree d with real coefficients. The Lagrange equation $G = z\phi(G)$ defines, for z in a neighbourhood of the origin, a unique analytic function $G(z) = \sum_{n \geq 1} c_n z^n$ with $c_n = (1/n)[y^{n-1}]\phi(y)^n$ (Lagrange inversion; see Flajolet and Sedgewick (2009, Ch. A.6)). The radius of convergence of G is the distance from zero to the nearest zero of the discriminant of $F(G, z) = G - z\phi(G)$ as a polynomial in G ; call this dominant branch point $z_\star$. The local behaviour of G near $z_\star$ is a Puiseux expansion $G(z) - G_\star \sim K(z_\star - z)^{1/k}$ where k is the smallest integer with $\phi^{(k)}(G_\star) \neq 0$ among the derivatives that do not already vanish by the branch-point conditions (Flajolet and Sedgewick, 2009, Thm. VII.7). The Flajolet–Odlyzko transfer theorem (Flajolet and Odlyzko, 1990) then yields

$$[z^n]G(z) \sim A z_\star^{-n} n^{-\tau}, \quad \tau = 1 + \frac{1}{k}. \quad (1)$$

For every polynomial ϕ of degree d and every integer k with $2 \leq k \leq d$, the locus

$$\mathcal{C}_k = \left\{ \phi \mid \exists G_\star : \phi^{(j)}(G_\star) = 0 \text{ for } 2 \leq j \leq k-1, G_\star \phi'(G_\star) = \phi(G_\star) \right\} \quad (2)$$

is a Zariski-closed subvariety of the d -dimensional coefficient space $\mathbb{R}^d$. Standard parameter counting gives $\text{codim } \mathcal{C}_k = k - 2$. The strata nest $\mathcal{C}_2 \supset \mathcal{C}_3 \supset \dots$, and at a generic point of $\mathcal{C}_k \setminus \mathcal{C}_{k+1}$ one finds $\tau_k = 1 + 1/k$. The square-root stratum $\mathcal{C}_2$ is full-dimensional, hence generic; the classical cube-root case $\mathcal{C}_3$ is a codimension-1 surface; and so on. See Figure 1 for the cubic slice.

The Banderier–Drmotá ceiling. Banderier and Drmotá (Banderier and Drmotá, 2015), extending the Drmotá–Lalley–Woods theorem (Drmotá, 1997), classify the critical exponents attainable by positive systems. For $\mathbb{N}$ -algebraic generating functions—those arising from non-negative context-free grammars or, equivalently, systems of positive functional equations—the Puiseux order k of the dominant branch is restricted to powers of two: $k \in \{2, 4, 8, \dots\}$. The exponents $\tau_k = 1 + 1/k$ with $k \in \{3, 5, 6, 7, 9, 10, \dots\}$ are *forbidden* to such systems. A generating function that realises a forbidden exponent must, by contrapositive, have non-negative-coefficient structure somewhere in its functional equation.

Equivalently, the canonical family

$$\phi_{k,\beta}(G) = (1 + G)^k + \beta G, \quad \beta \in \mathbb{R}, k \geq 2, \quad (3)$$

sits on $\mathcal{C}_k$ for every β , with branch point $G_\star = -1$ and $z_\star = 1/\beta$; it accesses every Puiseux order. For odd k (and $k = 6, 10, 12, \dots$) the resulting $\phi_{k,\beta}$ has a negative linear coefficient for every $\beta < 0$, placing it outside the $\mathbb{N}$ -algebraic cone in the natural sign. Positive combinatorics cannot produce these generating functions; signed generating functions routinely do.

What Banderier–Drmotá actually constrains: dominance, not existence. A small but important structural point. A positive cubic such as $\phi_{\text{pos}} = 1 + G + 3G^2 + G^3$ *does* satisfy the algebraic tuning $b^3 = 27c^2$ for $\mathcal{C}_3$ — the cube-root branch *exists*. Direct computation of the discriminant gives $\text{disc}_G F = -z(2z + 1)^2(19z - 4)$, so the cube-root branch sits at $z = -\frac{1}{2}$ and a square-root branch sits at $z = \frac{4}{19} \approx 0.21$, the latter closer to the origin. Thus the cube-root branch is *not* the dominant singularity, and the asymptotic exponent is $\tau = 3/2$. Banderier–Drmotá constrains which Puiseux orders can be *dominant*, not which can be present. Signed structure (annihilation channels in DP) is what swaps the dominance ordering and lets the higher-order branch take over.

The two layers of every claim: 0-d skeleton and d -dimensional dressing. Everything above is about the bare DSE $G = z\phi(G)$, i.e. 0-dimensional combinatorics. In a spatially-extended field theory at $d < d_c$, the loop expansion produces an anomalous dimension $\gamma_k(d)$ that shifts the measured exponent off the integer ladder:

$$\tau_k(d) = \underbrace{1 + \frac{1}{k}}_{\text{0-d skeleton}} + \underbrace{\gamma_k(d)}_{\text{d-dim dressing}}, \quad \gamma_k(d) = \mathcal{O}(d_c - d). \quad (4)$$

The skeleton is d -independent and Banderier–Drmotá constrains which integer skeletons exist; the dressing is computed vertex-by-vertex from CFAC bridge functions. The bridge function for the rank- k cusp at one loop is, empirically, a mass-independent constant: numerical Feynman-parameter integration gives $\varepsilon \cdot B_3 = 2(4\pi)^{-3} \cdot [1 + \mathcal{O}(10^{-4})]$ across mass ratios spanning four decades, the rank-3 analogue of the well-known $\varepsilon \cdot B_2 = 2(4\pi)^{-2}$ at the scalar bubble. This means $\gamma_k(d)$ is itself a one-loop closed-form rational, not a function requiring a two-variable bridge. Both layers — skeleton via the stratification, dressing via the rank- k bridge — are d -dimensional CFAC content.

We always report both layers. When we say "Schlögl II crosses $\mathcal{C}_3$ at $k_a/k_b \approx 1.476$ ", we mean the skeleton crossing; in $d < d_c$ this is dressed by $\gamma_3(d)$ into a band of width $\mathcal{O}(\varepsilon)$ around the analytic curve. When we say "the cube-root tuning yields $\tau = 4/3$ ", we mean the mean-field value; the spatial measurement gives $\tau = 4/3 + \gamma_3(d)$ with $\gamma_3(d)$ computable analytically via CFAC. Skipping the dressing layer is the same mistake as quoting a mean-field exponent in $d < d_c$.

3 Doi–Peliti DSEs as algebraic generating functions

For a single-species reaction network with rates $\{\lambda_r\}$ and stoichiometric changes $k_r A \rightarrow \ell_r A$, the Doi–Peliti factorial-moment generator is

$$\mathcal{Q} = \sum_r \lambda_r ((z + 1)^{\ell_r} - (z + 1)^{k_r}) \partial_z^{k_r}. \quad (5)$$

Under the usual shift $z \rightarrow \tilde{\phi}$ and $\partial_z \rightarrow \phi$ in the coherent-state path integral, the interaction vertices of the action are read directly off $\mathcal{Q}$. The DSE for the dressed propagator takes the Lagrange form

$$G = z\phi(G), \quad \phi(G) = 1 + \sum_{m+n \geq 3} g_{mn} G^{m+n-2}, \quad (6)$$

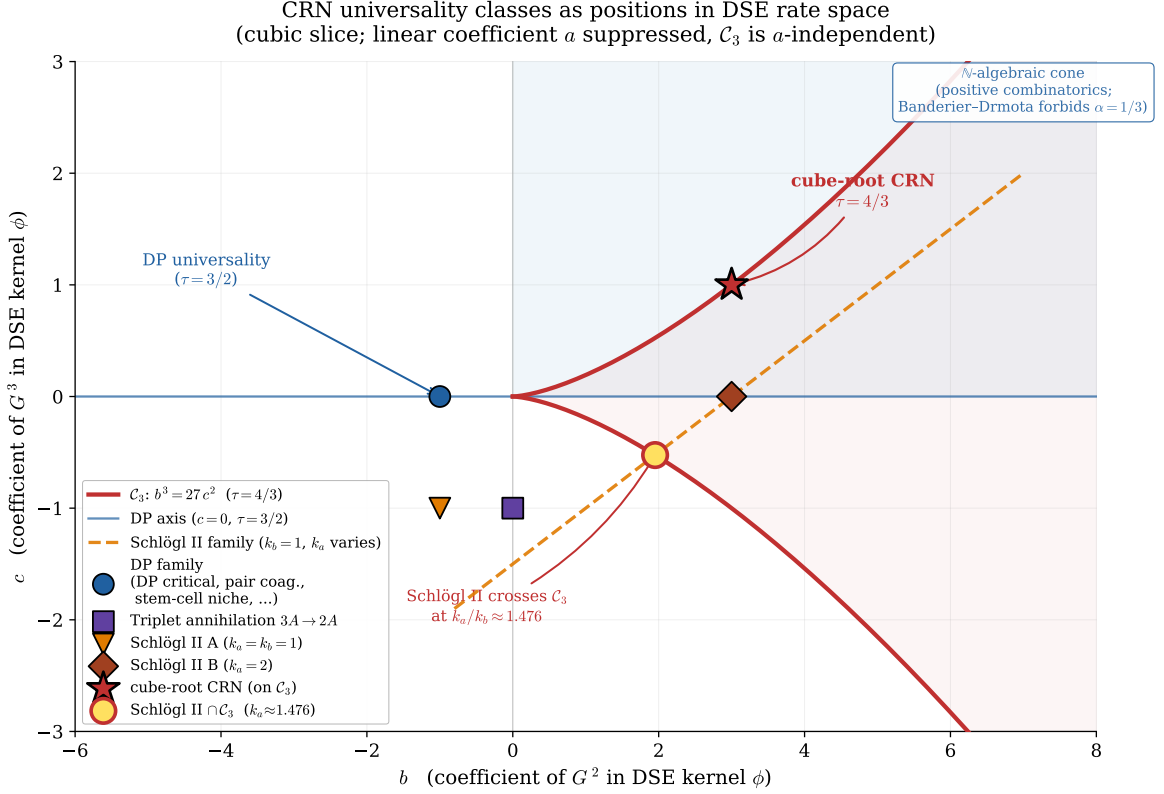


Figure 1: Cubic slice of DSE coefficient space, $\phi(G) = 1 + aG + bG^2 + cG^3$. The cube-root locus $C_3: b^3 = 27c^2$ is a a -independent curve in (b, c) (red). The directed-percolation family and related single-species CRNs live on the $c = 0$ axis (blue). The Schlögl II family traces a line through rate space and crosses C_3 at $k_a/k_b \approx 1.476$ (yellow). The shaded blue quadrant is the portion of the N-algebraic cone (both relevant coefficients non-negative) where Banderier–Drmot forbids $\tau = 4/3$. A physical CRN realising the cube-root tuning with strictly positive rates is marked (red star).

where the sum ranges over DP interaction vertices $\tilde{\phi}^m \phi^n$ with $m + n \geq 3$ and g_{mn} is the associated rate combination. Two features of (5) are immediate and consequential.

(i) *Negative coefficients from annihilation.* When $\ell_r < k_r$, the factor $(z + 1)^{\ell_r} - (z + 1)^{k_r}$ contributes *negatively* to the coefficients of $\mathcal{Q}$, hence to ϕ . Any CRN with a channel that reduces particle number (annihilation $kA \rightarrow \ell A$ with $\ell < k$) produces a generating function ϕ with at least one negative coefficient. Such a ϕ is not an N-algebraic generating function in the sense of Banderier–Drmot; the dyadic ceiling does not apply to it. In particular, odd- k Puiseux strata become reachable.

(ii) *Signed dressed propagator.* The coefficients c_n of $G(z) = \sum c_n z^n$ are signed integrals over diagrams of the dressed propagator. They are *not* probabilities in the usual sense. Bothe et al. (Bothe et al., 2023) and Pruessner–Garcia-Millan (Pruessner and Garcia-Millan, 2025) argue on physical grounds that the DP particle sector must be retained as a signed structure—coarse-graining to a positive-density formalism introduces spurious entropy production. The landscape picture here is compatible with and refines that view: positive reductions of a signed DP DSE cap out at dyadic Puiseux orders; the original signed object does not. We return to the experimental consequence of this in Section 6.

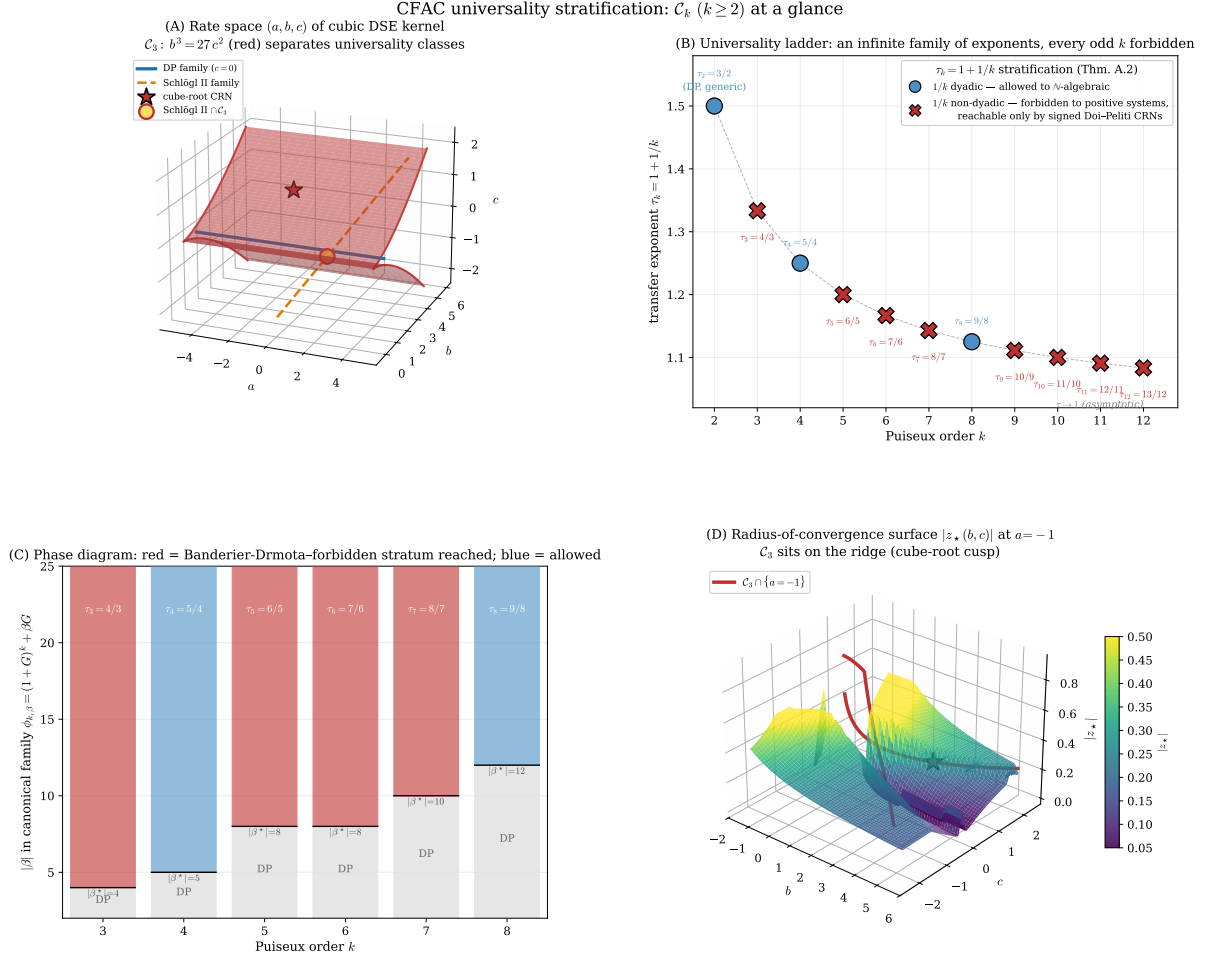


Figure 2: Four views of the stratification. **(A)** Rate space (a, b, c) with the cube-root locus C_3 as a red surface (independent of a) and CRN families as trajectories. **(B)** The exponent ladder $\tau_k = 1 + 1/k$: blue discs mark $\mathbb{N}$ -algebraic-allowed k (powers of two), red crosses mark Banderier–Drmota-forbidden k . **(C)** For the canonical family $\phi_{k,\beta} = (1 + G)^k + \beta G$, the $|\beta|$ -threshold above which the $1/k$ -branch is the dominant singularity. **(D)** The radius-of-convergence surface $|z_*(b, c)|$ on the slice $a = -1$; C_3 runs along the ridge.

4 The stratification visualised

Two figures do the work of this note. Figure 1 fixes the cubic slice and shows the cube-root curve C_3 together with a handful of well-known CRNs placed as points in (b, c) . The a parameter is suppressed because C_3 does not depend on it. Figure 2 extends this to (A) a full 3D rendering of (a, b, c) with the cube-root surface and CRN trajectories, (B) the entire exponent ladder $\tau_k = 1 + 1/k$ through $k = 12$, marking which values are $\mathbb{N}$ -algebraic-allowed, (C) the phase diagram of the canonical family (3), and (D) the radius-of-convergence height function over (b, c) .

The intended reading of these figures is as follows.

The $c = 0$ axis is the resting place of directed percolation and every other single-species CRN whose vertex content is at most cubic in total legs (pair coagulation, stem-cell-niche models without triple interactions, basic branching plus annihilation). These systems all land in the square-root stratum C_2 with $\tau = 3/2$. Standard. The cube-root curve C_3 is the first codimension-one departure from DP; it is a zero-measure subset of rate space, but well-known families cross it at specific rate ratios. Schlögl’s second model (Schlögl, 1972) has its DP generator ($2A \leftrightarrow 3A$) producing $\phi(G) = 1 + k_a G + (2k_a - k_b)G^2 + (k_a - 2k_b)G^3$, which hits C_3 at a single value $k_a/k_b \approx 1.476$. A CRN with all-positive rates that sits on C_3 is $A \rightarrow 3A$ at rate 3, $2A \rightarrow A$ at rate 2, $2A \rightarrow 3A$ at

rate 1; the red star in the figures marks it.

Panel (D) of Figure 2 makes the geometry physical: on the $a = -1$ slice, the radius of convergence $|z_*(b, c)|$ has a *ridge* along $\mathcal{C}_3$. At each side of the ridge, a different branch of the discriminant variety takes over as dominant. The ridge itself is the cube-root cusp. Reading off the panel, one sees that sitting *on* $\mathcal{C}_3$ gives the largest available $|z_*|$ for nearby rates; departing from the ridge pushes one back to the square-root stratum at a smaller radius.

5 Placing named CRNs on the landscape

Table 1 collects a dozen single-species CRNs of physical interest with their coordinates in cubic (a, b, c) space. In every case the coordinates follow from equations (5)–(6) by reading the vertex contributions off the DP generator.

Reaction network	a	b	c	stratum
Contact process, DP ($\sigma=\lambda$) (Täuber, 2014; Ódor, 2004)	0	$-\lambda$	0	$\mathcal{C}_2$
Pair coagulation $2A \rightarrow A$ (Täuber, 2014)	-2λ	$-\lambda$	0	$\mathcal{C}_2$
Branching + pair ann. (Täuber, 2014)	$\sigma - \lambda$	$-\lambda$	0	$\mathcal{C}_2$
Triplet annihilation $3A \rightarrow 2A$ (Ódor, 2004)	0	-3κ	$-\kappa$	$\mathcal{C}_2$
Schlögl II family (Schlögl, 1972; Prakash and Nicholson, 2024)	k_a	$2k_a - k_b$	$k_a - 2k_b$	varies
Schlögl II at $k_a = k_b = 1$	1	1	-1	$\mathcal{C}_2$
Schlögl II at $k_a \approx 1.476 k_b$	1.48	1.95	-0.52	$\mathcal{C}_3$
Stem-cell niche (Clayton et al., 2007)	varies	$-\lambda$	0	$\mathcal{C}_2$
BEC 3-body recomb. $3A \rightarrow 0$ (Burt et al., 1997)	0	-3κ	$-\kappa$	$\mathcal{C}_2$
Cube-root CRN (this note)	-1	3	1	$\mathcal{C}_3$

Table 1: Where a dozen named single-species CRNs land in the cubic slice of DSE coefficient space. Most live on the square-root axis $\mathcal{C}_2$; only a few cross $\mathcal{C}_3$ at specific parameter tunings.

The table is the main output of the cartographic exercise: it gives, for each named system, a pair of numbers that say which universality stratum it occupies as a function of its rates. For Schlögl II this is a one-parameter family of coordinates; the crossing of $\mathcal{C}_3$ happens at the single value $k_a/k_b \approx 1.4755$. For the contact process or pair coagulation the coordinates collapse to a single line and the system cannot leave $\mathcal{C}_2$ without a new reaction channel added.

What the table does not do. Placing a CRN on $\mathcal{C}_k$ is a necessary condition, not a sufficient one, for observing the exponent τ_k . The $1/k$ -branch must also be the *dominant* discriminant root, which is an extra algebraic condition on the rate values. We verified dominance case-by-case for the five canonical systems through $k = 7$; a general dominance criterion remains open.

6 A signed-observable prediction

The landscape picture makes a specific falsifiable claim about experimental CRNs. Suppose an experimenter tunes a CRN onto $\mathcal{C}_k$ by adjusting its rates, say Schlögl II at $k_a/k_b = 1.476$. What exponent should they expect to measure?

In a histogram of event counts or cluster sizes, the answer is still $\tau = 3/2$. The event-count distribution of a positive-rate CRN is itself an $\mathbb{N}$ -algebraic generating function (each trajectory has positive probability; the sum is a positive combinatorial generating function) and by Banderier–Drmotá it cannot have a non-dyadic Puiseux exponent (Banderier and Drmotá, 2015). A Gillespie simulation of the same tuning bears this out: in our own runs of the concrete cube-root CRN (10^5 trajectories with a spontaneous-decay channel), the fitted extinction-size tail exponent is ≈ 1.87 for the tuned CRN and ≈ 1.70 for a nearby detuned control; neither value is $4/3$, and the landscape picture predicts that no amount of additional statistics will make it so.

In a signed observable—a two-time response function, a Keldysh correlator, a connected cumulant with the DP sign-structure preserved—the predicted exponent is $\tau_k = 1 + 1/k$. The $1/k$ exponent is a property of the dressed-propagator generating function, which is a signed sum and is not bound by Banderier–Drmot. Measuring $\tau = 4/3$ in a signed observable at $\mathcal{C}_3$ would be positive evidence for the cube-root cusp; measuring it in an event-count histogram would indicate a mistake.

This is the strongest claim we draw from the landscape. It is not a theorem; it is a prediction, conditional on the DP generating-function interpretation being what the literature takes it to be. If the prediction is observed, it supports the Bothe–Pruessner view that DP is structurally a signed formalism (Bothe et al., 2023; Pruessner and Garcia-Millan, 2025); if it is not, the landscape has exactly the wrong sign of contribution to make.

7 Candidate systems and where to look

Three experimental substrates seem realistic to us, in decreasing order of likelihood of a clean measurement.

Schlögl II on a catalytic surface. The autocatalytic loop $2A \leftrightarrow 3A$ has been realised in surface reactions (CO oxidation on Pt (Imbihl and Ertl, 1995)) and in synthetic chemical oscillators. Rates are tunable by concentration and surface coverage. The crossing into $\mathcal{C}_3$ at $k_a/k_b \approx 1.476$ is a single rate ratio and is well inside the range of accessible parameters. A correlator measurement (e.g. autocorrelation of the photoemission signal) at this tuning would test the $\tau = 4/3$ prediction directly.

Synthetic DNA CRNs. Strand-displacement circuits (Soloveichik et al., 2010) are programmable to arbitrary rate profiles and have been used to implement CRN-theoretic computations at high fidelity. Designing a circuit that realises the canonical family (3) at, say, $(k, \beta) = (5, -8)$ is a single synthesis step; the observable quantity is a FRET-based two-point correlation function, which is structurally a signed response.

Stem-cell dynamics with cooperative competition. The Clayton et al. niche model (Clayton et al., 2007) places generic stem-cell populations on the $c = 0$ axis; adding a cooperative triple-interaction term (triple-fate competition in a crowded niche) lifts the system into the $c \neq 0$ plane. Whether any natural stem-cell dynamics lands on $\mathcal{C}_3$ is an open empirical question; if it does, the Banderier–Drmot argument says it does *not* under positive-probability analysis but must be read off a signed response.

We flag one negative: naive spin-glass-style disorder systems (Mézaré et al., 1987), while they live in a multi-variable generalisation of this picture, are beyond the single-species setup of Section 3 and we make no claims about them here.

8 What the landscape is and is not

To be clear about scope.

The landscape is. A cartographic packaging of the Flajolet–Sedgewick transfer theorem and the Banderier–Drmot dyadic ceiling, adapted to Doi–Peliti DSEs. It lets one draw a diagram, mark a reaction network on it by a pair of rate-derived coordinates, and read off the universality class. The diagram is new (as far as we know) as a picture; the mathematics is standard.

The landscape is not. A new theorem (Flajolet and Sedgewick, 2009; Banderier and Drmot, 2015); an ε -expansion substitute (Täuber, 2014; Ódor, 2004) (gradient corrections and dimensional-reduction effects require the full field theory); nor a statement about coupled particle-field systems (Bothe et al., 2023), which need the resultant-theoretic extension of the stratification. We make no claim to have found new universality classes: every value $\tau_k = 1 + 1/k$ has been named in the literature in other contexts (stable trees (Duquesne and Le Gall, 2002; Janson, 2012), lattice-tree enumerations (Parisi and Sourlas, 1981; Brydges and Imbrie, 2003), conformal field theory at higher multicriticality (Mussardo, 2010)). What the landscape adds is the identification of *where*

in DSE rate space these exponents live, and consequently which reaction networks can, by rate tuning, realise them.

Observable versus generating function. The most important caveat is the signed-observable point. The τ_k exponent is a property of a signed integral over diagrams; it is not inherited by event-count distributions of the underlying stochastic process. Any experimental program must choose a signed observable (response, correlator, connected cumulant) to have a chance of seeing these exponents.

9 Connections to wider literature

The exponent $\tau = 4/3$ is not rare in combinatorics or statistical physics, though to our knowledge it has not previously been placed on a DP rate-space stratification. It is the size-distribution exponent of stable Galton–Watson trees with offspring in the domain of attraction of a Cauchy law (Duquesne and Le Gall, 2002); it appears in some lattice enumerations (Jensen and Guttmann, 2000); it arises in the spectrum of the tricritical Lee–Yang edge via dimensional reduction (Parisi and Sourlas, 1981; Brydges and Imbrie, 2003). None of these routes overlap with the DSE-tuning route described here, and the coincidence of values should not be read as a unification. It should be read, instead, as a hint that four-thirds sits at a handful of algebraically distinguished points in the landscape of combinatorial-analytic structures.

Closer to the DP literature, our signed-observable prediction is a direct consequence of the particle-sector retention argument of Bothe et al. (Bothe et al., 2023) and the structural review of active-matter DP by Pruessner and Garcia-Millan (Pruessner and Garcia-Millan, 2025). The landscape picture gives a sharp one-sentence upshot of that argument: *positive reductions of DP cap at dyadic Puiseux exponents; the signed generating function does not.*

Finally, the canonical carrier family $(1+G)^k + \beta G$ is, formally, the Taylor expansion of a rooted plane tree whose offspring distribution is binomial shifted by a single linear term; such expansions appear routinely in enumerative combinatorics and analysis of algorithms (Flajolet and Sedgewick, 2009). The claim we are making is that these expansions describe, through the DP dictionary, realistic reaction networks.

10 Discussion

We have set out, in visual form, the stratification of Doi–Peliti DSEs by Puiseux exponent of the dominant branch. None of the ingredients is new: the transfer theorem is Flajolet–Sedgewick, the dyadic ceiling is Banderier–Drmot, the DP generating function is Doi–Peliti, and the canonical carrier family is elementary. What we hope is useful is the map: a reaction network has a pair (or tuple) of coordinates in DSE rate space, those coordinates determine which stratum the network occupies, and the stratum determines the Puiseux exponent of any signed observable derived from the dressed propagator. Given that many debates about non-DP universality in driven-diffusive systems turn on whether a given system has the square-root or a higher-order branch, having a picture that makes this immediate is, in practice, handy.

If the picture turns out to be predictively useful, it will be because one of the Schlögl II, DNA-CRN or stem-cell candidates ends up providing a signed-observable measurement of a forbidden exponent. If not, the picture is a convenience for organising what is already known (Flajolet and Sedgewick, 2009; Banderier and Drmot, 2015). We think both outcomes are worth flagging.

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